

Dear Editor,

Please consider our manuscript, “Remote sensing large-wood storage downstream of reservoirs during and after dam removal: Elwha River, Washington, USA”, for possible publication in *Earth and Space Science*.

The manuscript outlines several methodological advances in the accurate and automated quantification of large wood in rivers from remotely sensed imagery and presents a detailed application case study to demonstrate the usefulness of these new comprehensive measurements. We show, for the first time, that such measurements can reveal the relative timing and magnitude of large wood and sediment deposition that result from dam removal. Specifically, we examine the change in abundance and dynamics of large wood and sediment due to the largest dam removal to date, on the Elwha River in Washington State, USA, which is the first such study on this or any other river system.

We are keen to publish this work in *Earth and Space Science* for two principal reasons. The first is the commitment to open science, because our work is heavily underpinned by the widespread availability of our data and models, which we claim to be of benefit to the large wood in rivers research community. Second, the current manuscript builds upon previous methodological advances published in *Earth and Space Science*, specifically related to image labeling and segmentation using Machine Learning (Buscombe et al., 2022; Buscombe and Goldstein, 2022), as well as previous applications of similar techniques for other comprehensive measurement targets in rivers (Bodine et al., 2022).

This work has not been submitted for review elsewhere. The authors have no conflicts of interest to declare.

Sincerely,

Daniel Buscombe and co-authors

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References

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